

Randomized Phase III KEYNOTE-181 Study of Pembrolizumab Versus Chemotherapy in Advanced Esophageal Cancer

Takashi Kojima, MD¹; Manish A. Shah, MD²; Kei Muro, MD³; Eric Francois⁴; Antoine Adenis, MD, PhD⁵; Chih-Hung Hsu, MD, PhD⁶; Toshihiko Doi, MD, PhD⁷; Toshikazu Moriwaki, MD, PhD⁸; Sung-Bae Kim, MD, PhD⁹; Se-Hoon Lee, MD, PhD¹⁰; Jaafar Bennouna, MD, PhD¹¹; Ken Kato, MD, PhD¹²; Lin Shen, PhD¹³; Peter Enzinger, MD¹⁴; Shu-Kui Qin, MD¹⁵; Paula Ferreira¹⁶; Jia Chen, PhD¹⁷; Gustavo Giroto, MD¹⁸; Christelle de la Fouchardiere, MD¹⁹; Helene Senellart, MD²⁰; Raed Al-Rajabi, MD²¹; Florian Lordick²²; Ruixue Wang, PhD²³; Shailaja Suryawanshi, PhD²³; Pooja Bhagia, MD²³; S. Peter Kang, MD²³; and Jean-Philippe Metges²⁴ on behalf of the KEYNOTE-181 Investigators

abstract

PURPOSE Patients with advanced esophageal cancer have a poor prognosis and limited treatment options after first-line chemotherapy.

PATIENTS AND METHODS In this open-label, phase III study, we randomly assigned (1:1) 628 patients with advanced/metastatic squamous cell carcinoma or adenocarcinoma of the esophagus, that progressed after one prior therapy, to pembrolizumab 200 mg every 3 weeks for up to 2 years or chemotherapy (investigator's choice of paclitaxel, docetaxel, or irinotecan). Primary end points were overall survival (OS) in patients with programmed death ligand-1 (PD-L1) combined positive score (CPS) ≥ 10 , in patients with squamous cell carcinoma, and in all patients (one-sided α 0.9%, 0.8%, and 0.8%, respectively).

RESULTS At final analysis, conducted 16 months after the last patient was randomly assigned, OS was prolonged with pembrolizumab versus chemotherapy for patients with CPS ≥ 10 (median, 9.3 v 6.7 months; hazard ratio [HR], 0.69 [95% CI, 0.52 to 0.93]; $P = .0074$). Estimated 12-month OS rate was 43% (95% CI, 33.5% to 52.1%) with pembrolizumab versus 20% (95% CI, 13.5% to 28.3%) with chemotherapy. Median OS was 8.2 months versus 7.1 months (HR, 0.78 [95% CI, 0.63 to 0.96]; $P = .0095$) in patients with squamous cell carcinoma and 7.1 months versus 7.1 months (HR, 0.89 [95% CI, 0.75 to 1.05]; $P = .0560$) in all patients. Grade 3-5 treatment-related adverse events occurred in 18.2% of patients with pembrolizumab versus 40.9% in those who underwent chemotherapy.

CONCLUSION Pembrolizumab prolonged OS versus chemotherapy as second-line therapy for advanced esophageal cancer in patients with PD-L1 CPS ≥ 10 , with fewer treatment-related adverse events.

J Clin Oncol 38:4138-4148. © 2020 by American Society of Clinical Oncology

INTRODUCTION

Esophageal cancer is the seventh most common cancer and the sixth most common cause of cancer-related death worldwide, with approximately 572,000 new cases and 509,000 deaths in 2018.¹ The highest prevalence of esophageal cancer occurs in Asia and Africa, where the most common subtype is squamous cell carcinoma, whereas adenocarcinoma is more common in North America and Western Europe.¹⁻³ Treatment options for patients with unresectable, locally advanced, or metastatic esophageal cancer are limited.⁴ The prognosis is typically poor in patients with metastatic esophageal cancer, with 5-year survival rates of less than 5%.⁵ Current guidelines recommend combination with fluoropyrimidine and platinum therapies in first-line chemotherapy.^{2,6,7} However, after first-

line chemotherapy, there is no accepted standard of care, although taxanes and irinotecan are used.⁸

Anti-programmed death 1 (PD-1)/programmed death ligand-1 (PD-L1) therapies have shown antitumor activity in patients with metastatic esophageal cancer.⁹⁻¹² Pembrolizumab is a humanized, high-affinity, monoclonal anti-PD-1 antibody that provides a survival benefit in multiple tumor types.¹³ In the phase II KEYNOTE-180 study (ClinicalTrials.gov identifier: NCT02559687), pembrolizumab provided durable responses with acceptable safety in patients with esophageal cancer who had progressed after two or more prior therapies.¹¹ Here, we report the results from the randomized phase III, open-label KEYNOTE-181 study of pembrolizumab versus paclitaxel, docetaxel, or irinotecan in patients with advanced or metastatic

ASSOCIATED CONTENT

Data Supplement Protocol

Author affiliations and support information (if applicable) appear at the end of this article.

Accepted on September 1, 2020 and published at ascopubs.org/journal/jco on October 7, 2020; DOI <https://doi.org/10.1200/JCO.20.01888>

CONTEXT

Key Objective

What is the antitumor activity of pembrolizumab versus chemotherapy as second-line treatment in patients with advanced or metastatic esophageal cancer?

Knowledge Generated

Pembrolizumab provided a clinically meaningful survival benefit versus chemotherapy for patients with metastatic esophageal squamous cell carcinoma and programmed death ligand-1 (PD-L1) combined positive score (CPS) ≥ 10 tumors and also in patients with metastatic esophageal squamous cell carcinoma or PD-L1 CPS ≥ 10 tumors, in the second-line, with reduced toxicity.

Relevance

Pembrolizumab provided a clinically meaningful overall survival benefit versus chemotherapy in a global population of patients with metastatic esophageal squamous cell carcinoma and with PD-L1 CPS ≥ 10 tumors in the second-line setting.

esophageal cancer who progressed after one line of prior therapy.

PATIENTS AND METHODS

Patients

Eligible patients were ≥ 18 years of age with histologically confirmed squamous cell carcinoma or adenocarcinoma of the esophagus including human epidermal growth factor receptor 2/neu negative Siewert type 1 adenocarcinoma of the esophagogastric junction. Selection criteria included metastatic or locally advanced, unresectable disease, measurable disease per RECIST version 1.1 by local investigator/radiology assessment, documented radiographic or clinical progression on one prior line of standard therapy, and Eastern Cooperative Oncology Group performance status of 0 or 1 (on a 5-point scale, with higher scores indicating increasing disability).¹⁴ Patients also had to provide a newly obtained or archival tissue sample, and written informed consent. Full eligibility criteria are presented in the Protocol (online only).

Trial Design and Treatment

In this randomized, open-label, global, phase III study, patients were randomly assigned 1:1 to pembrolizumab 200 mg every 3 weeks or investigator's choice of standard-of-care chemotherapy with paclitaxel 80-100 mg/m² on days 1, 8, and 15 of each 28-day cycle, docetaxel 75 mg/m² on day 1 of each 21-day cycle, or irinotecan 180 mg/m² on day 1 of each 14-day cycle. Random assignment was stratified by histology (squamous cell carcinoma v adenocarcinoma) and geographic region (Asia v rest of world). Treatment with pembrolizumab or paclitaxel, docetaxel, or irinotecan was continued until documented disease progression, unacceptable toxicity or physician or patient decision to withdraw, or after up to 2 years of

pembrolizumab. Additional study and treatment details are provided in the Protocol.

Assessments

PD-L1 expression was centrally assessed during screening using the PD-L1 IHC 22C3 pharmDx assay (Agilent Technologies, Carpinteria, CA). Tumors positive for PD-L1 had a combined positive score (CPS) of 10 or more as described previously.¹¹ Tumor response was assessed per RECIST version 1.1 by central radiology review at week 9 and every 9 weeks thereafter. Progressive disease was verified by central imaging review. During follow-up, survival was assessed every 9 weeks. Adverse events (AEs) were assessed throughout the study and at 30 days (90 days for serious AEs and events of interest to pembrolizumab) after treatment discontinuation and were graded according to National Cancer Institute Common Terminology Criteria for Adverse Events (version 4.0).

Trial Oversight

The study was designed by academic investigators and employees of Merck Sharp & Dohme Corp, a subsidiary of Merck & Co, Inc, Kenilworth, NJ. The Protocol was approved by the appropriate institutional review board or ethics committee at each participating institution. All authors attest that the trial was conducted in accordance with the Protocol and all its amendments and Good Clinical Practice standards. All authors had full access to the data and were involved in the writing or reviewing and editing drafts of the manuscript and vouch for the accuracy and completeness of the data analyses. Assistance in the preparation of the manuscript was provided by a medical writer employed by the sponsor.

End Points

The three primary end points were overall survival (OS) in patients with PD-L1 CPS ≥ 10 , in patients with squamous cell carcinoma, and in all patients. Secondary end points

included progression-free survival (PFS) and objective response rate (ORR) per RECIST version 1.1 by central review in patients with PD-L1 CPS ≥ 10 , in patients with squamous cell carcinoma, and in all patients, and safety and tolerability.

Statistical Analyses

The statistical analysis plan specified one interim analysis and a final analysis. The overall type-1 error rate was strongly controlled at a one-sided α of 2.5% with use of the graphical method of Maurer and Bretz (Data Supplement, online only). The Lan-DeMets O'Brien-Fleming α spending function was used to control for type 1 error. The secondary hypotheses of PFS and ORR in all patients were tested only if OS with pembrolizumab was superior to that with chemotherapy in all patients.

We determined that a global enrollment of 600 patients (280 with PD-L1 CPS ≥ 10 and 400 with squamous cell

carcinoma of the esophagus) would permit comparison of superiority for pembrolizumab versus chemotherapy in patients with PD-L1 CPS ≥ 10 , in patients with squamous cell carcinoma, or in all patients to have at least 90.9%, 91.3%, or 92.6% power with underlying hazard ratios (HRs) of 0.60, 0.65, or 0.70 for OS at a one-sided α level of 0.9%, 0.8%, or 0.8%, respectively. At the Protocol-specified final analysis (data cutoff date October 15, 2018), death events for two patients were not included in the data analysis because of a data reporting inconsistency. As such, a subsequent OS analysis was performed at the October 15, 2018, cutoff date to include the events for these patients. The OS results reported are based on the final analysis and the updated analysis that includes death events for these two patients. In addition, an OS analysis was performed with 4 months of additional follow-up (data cutoff date February 13, 2019). The initial multiplicity strategy is applicable only to the final

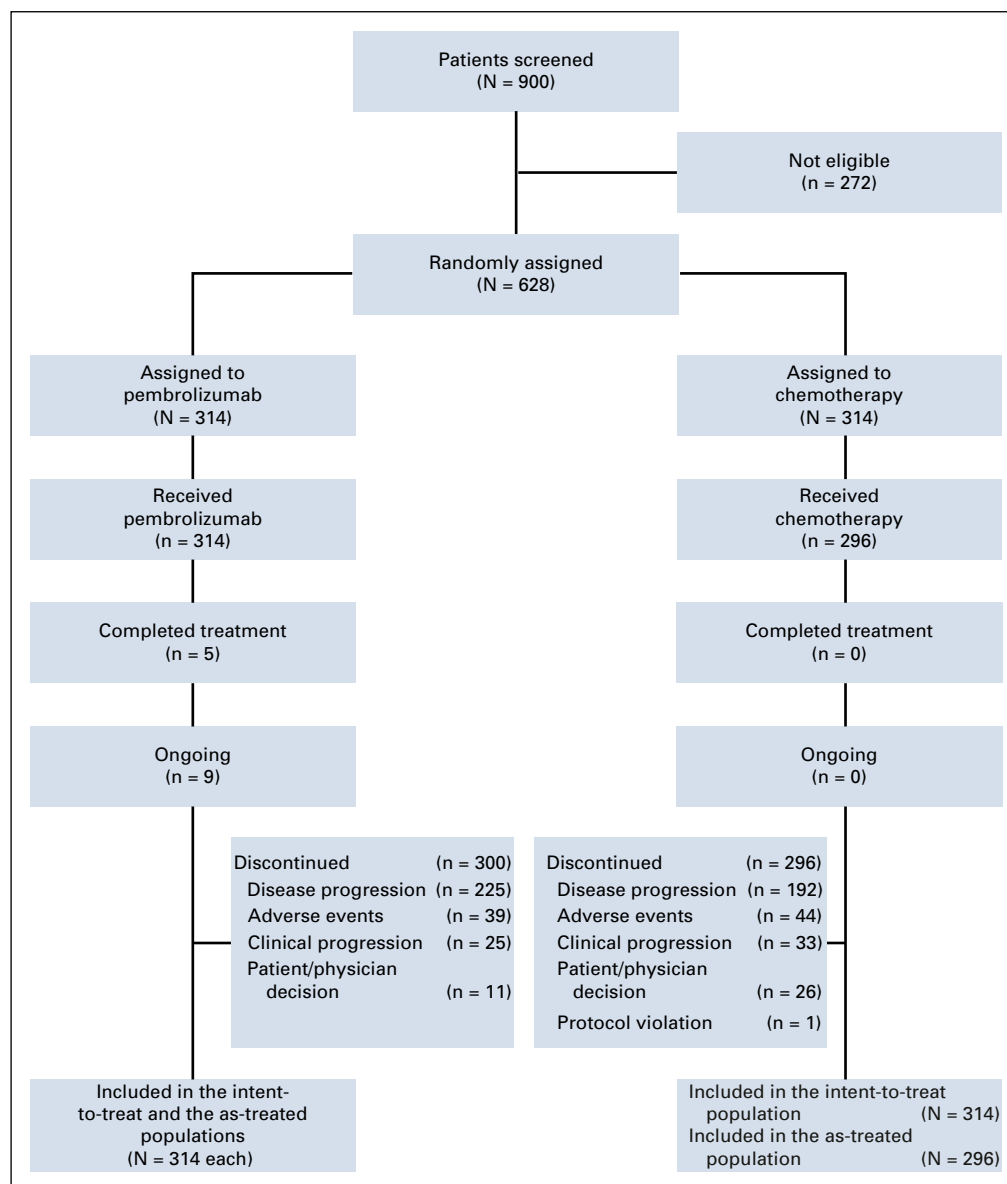


FIG 1. Study profile.

TABLE 1. Baseline Patient Demographic and Disease Characteristics

Characteristic	Pembrolizumab (N = 314)	Chemotherapy (N = 314)
Age, years, median (range)	63.0 (23-84)	62.0 (24-84)
≥ 65 years	139 (44.3)	133 (42.4)
Male	273 (86.9)	271 (86.3)
Geographic region		
Asia	121 (38.5)	122 (38.9)
Rest of world	193 (61.5)	192 (61.1)
ECOG performance status ^a		
0	126 (40.1)	116 (36.9)
1	187 (59.6)	197 (62.7)
2	1 (0.3)	1 (0.3)
Histology		
Squamous cell carcinoma ^b	198 (63.1)	203 (64.6)
Adenocarcinoma	116 (36.9)	111 (35.4)
PD-L1 combined positive score ^c		
≥ 10	107 (34.1)	115 (36.6)
< 10	201 (64.0)	196 (62.4)
Not evaluable ^d	6 (1.9)	3 (1.0)
Prior adjuvant or neoadjuvant therapy		
Yes	32 (10.2)	32 (10.2)
Disease stage		
Metastatic	290 (92.4)	286 (91.1)
Locally advanced	24 (7.6)	28 (8.9)
No. of prior therapies		
0	2 (0.6)	0 (0.0)
1	303 (96.5)	310 (98.7)
≥ 2	9 (2.9)	4 (1.3)

NOTE. Data are presented as No. (%) unless indicated otherwise. Patients in the chemotherapy group received investigator's choice of paclitaxel, docetaxel, or irinotecan. Percentages may not total 100 because of rounding. There were no significant differences between treatment groups at baseline. Data cutoff date was October 15, 2018.

Abbreviations: ECOG, Eastern Cooperative Oncology Group; PD-L1, programmed death ligand-1.

^aECOG performance status scores range from 0 to 5, with 0 indicating no symptoms and higher scores indicating greater disability.

^bAt the data cutoff date of February 13, 2019, the number of patients with squamous cell carcinoma increased to 199 (pembrolizumab) and 204 (chemotherapy).

^cThe PD-L1 combined positive score was defined as the number of PD-L1–positive cells (tumor cells, macrophages, and lymphocytes) divided by the total number of tumor cells.

^dPD-L1 expression could not be evaluated because samples had an inadequate number of cells or no cells.

analysis (Data Supplement). Precision to the fifth decimal place of the nonparametric rank-based test used to assess survival is limited because survival time is collected in ≥ 24-hour intervals. The updated analysis and the 4-month follow-up analysis are post hoc and are not subject to the multiplicity model.

RESULTS

Patients and Treatment

Between December 8, 2015, and June 16, 2017, a total of 628 patients from 154 sites in 32 countries were randomly

assigned to pembrolizumab (314 patients) or chemotherapy (314 patients; Fig 1). Baseline demographic and disease characteristics were as expected, with no significant differences between the groups (Table 1). Most patients (544 [86.6%]) were male, and 401 (63.9%) had squamous cell carcinoma. Expression of PD-L1 CPS ≥ 10 was well balanced between the groups, with 222 patients (35.4%) having PD-L1 CPS ≥ 10 (107 [34%] in the pembrolizumab group and 115 [37%] in the chemotherapy group). At the data cutoff date of October 15, 2018, the median follow-up duration from random assignment to data cutoff or death, whichever came first,

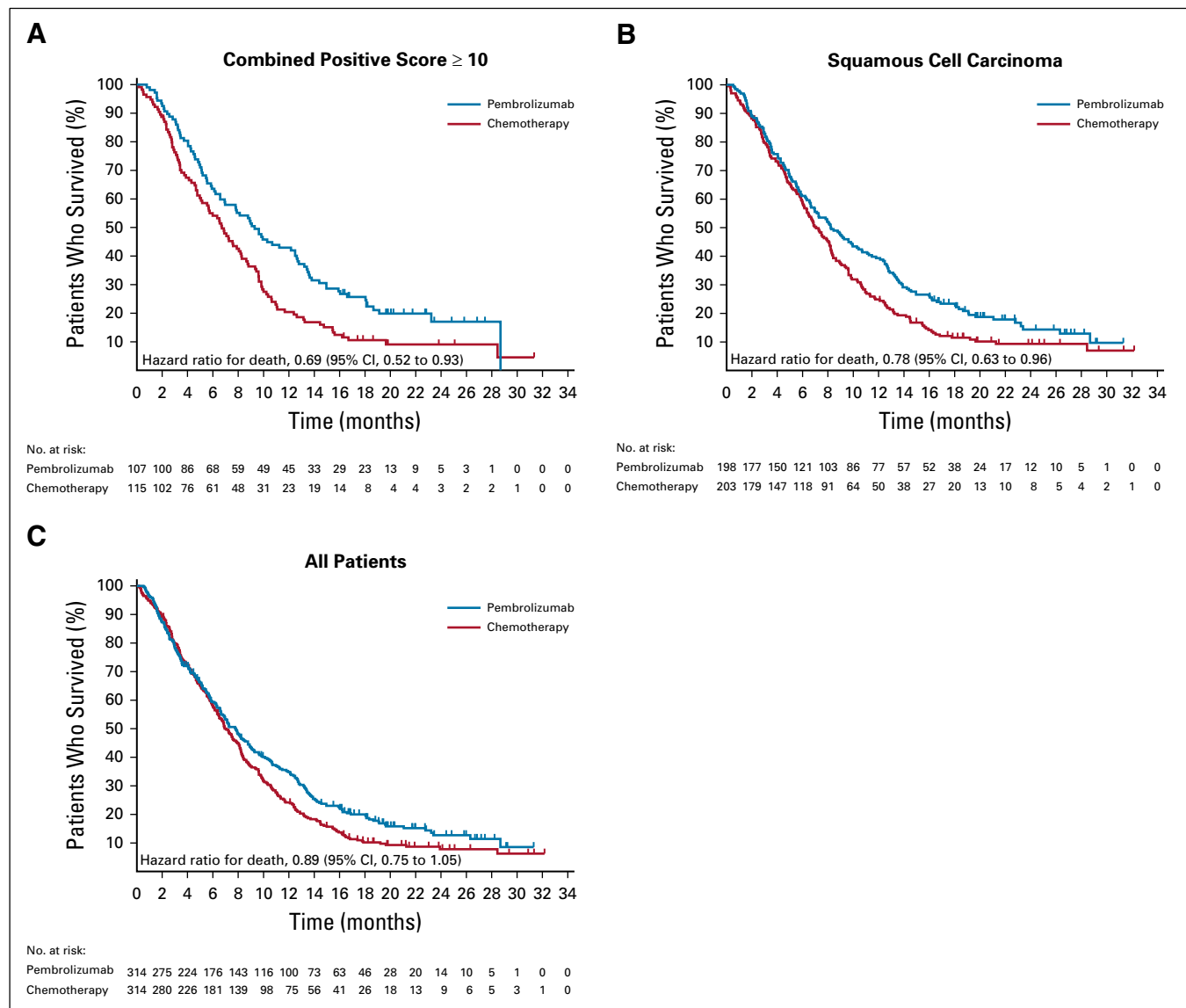
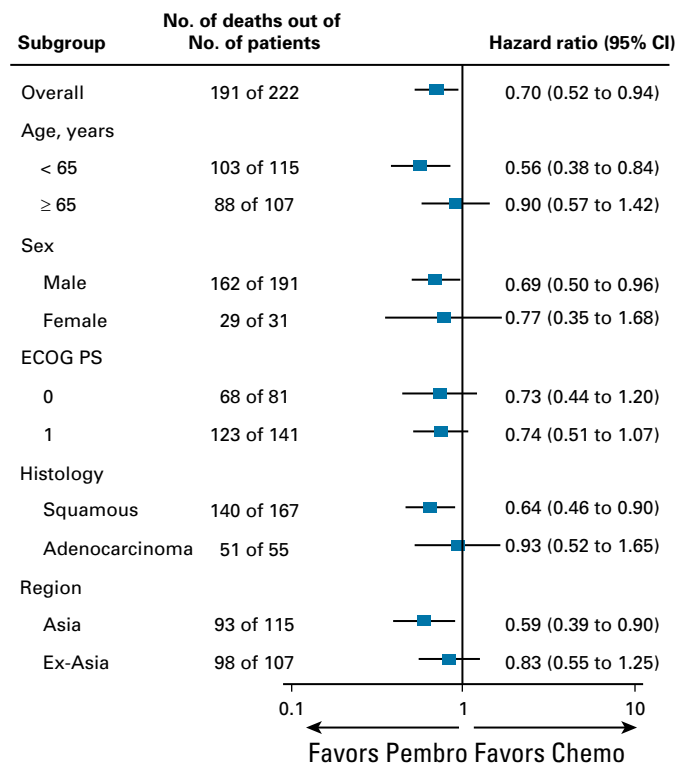
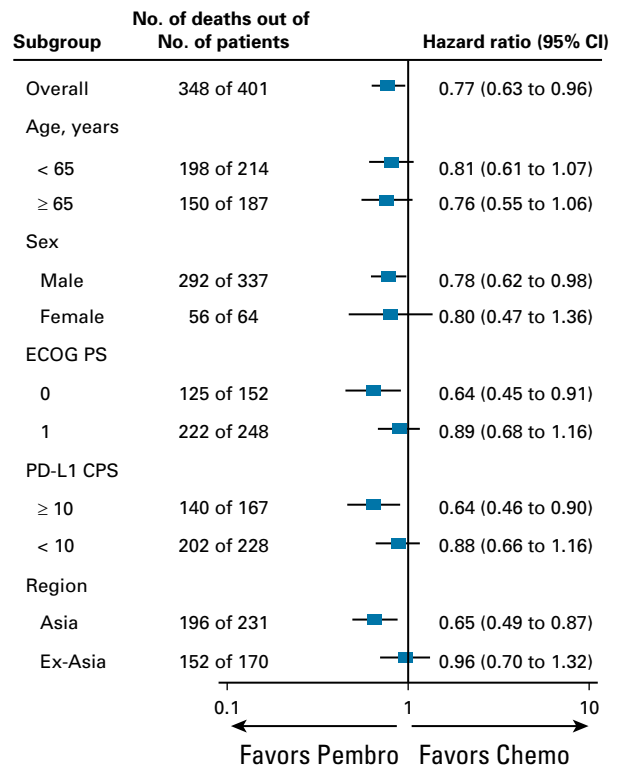
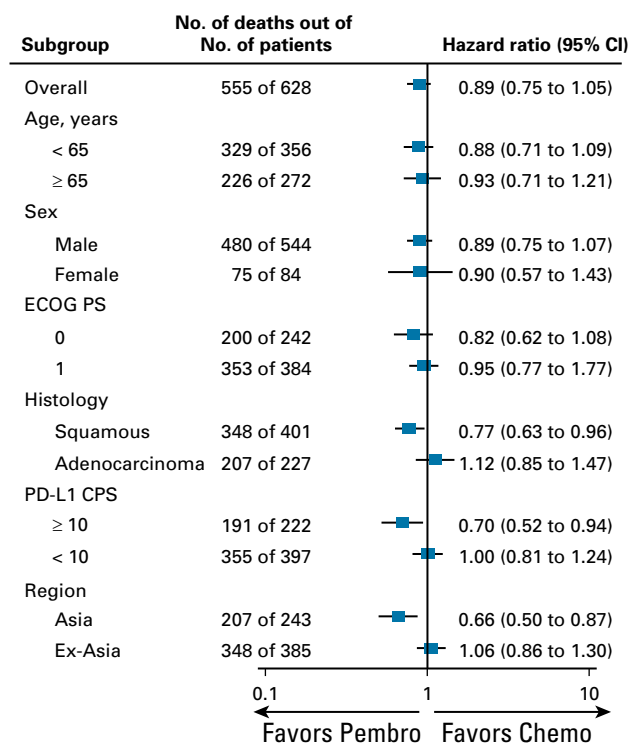


FIG 2. Overall survival in (A) patients with programmed death ligand 1 combined positive score ≥ 10 , (B) patient with squamous cell carcinoma, and (C) the intent-to-treat population. The data cutoff date of the original final analysis of overall survival was October 15, 2018.

was 7.1 months (range, 0.5-31.3 months) for patients in the pembrolizumab group and 6.9 months (range, 0.2-32.2 months) in the chemotherapy group. A total of five patients (1.6%) completed 2 years of therapy (four with squamous cell carcinoma [PD-L1 CPS < 10] and one with adenocarcinoma [PD-L1 CPS ≥ 10]), and nine patients (2.9%) continued to receive therapy (five with squamous cell carcinoma [four PD-L1 CPS ≥ 10 and one PD-L1 CPS < 10] and four with adenocarcinoma [one PD-L1 CPS ≥ 10 and three PD-L1 CPS < 10]) in the pembrolizumab group; no patients continued to receive therapy in the chemotherapy group (Fig 1). A total of 31 patients (5%; one in the pembrolizumab group and 30 in the chemotherapy group) received treatment after progression with checkpoint inhibitor therapies.

OS

At final analysis, a total of 190 patients with PD-L1 CPS ≥ 10 had died (87 [81.3%] in the pembrolizumab group and 103 [89.6%] in the chemotherapy group). Median OS was 9.3 months (95% CI, 6.6 to 12.5 months) and 6.7 months (95% CI, 5.1 to 8.2 months), respectively (Data Supplement). This difference met the prespecified boundary for showing superiority ($P < .00853$) of OS with pembrolizumab versus chemotherapy (HR, 0.69 [95% CI, 0.52 to 0.93]; $P = .0074$; Fig 2). The 12-month survival rate was 43.0% in the pembrolizumab group and 20.4% in the chemotherapy group (Data Supplement). The updated analysis showed that 191 patients with PD-L1 CPS ≥ 10 had died (88 [82.2%] in the pembrolizumab group and 103 [89.6%] in the chemotherapy group). The HR was 0.70

A
Combined Positive Score ≥ 10

B
Squamous Cell Carcinoma

C
All Patients


(95% CI, 0.52 to 0.94) for pembrolizumab versus chemotherapy; this difference did not meet the prespecified boundary for showing superiority, $P < .00855$ (Data Supplement). With an additional 4 months of follow-up, the HR was 0.67 (95% CI, 0.50 to 0.89; Data Supplement). Survival outcomes were generally similar with pembrolizumab versus chemotherapy across key prespecified subgroups of patients with CPS ≥ 10 , with the greatest survival benefit observed in patients with squamous cell carcinoma and in those from Asia (Fig 3).

The study did not meet the coprimary end point of OS in patients with squamous cell carcinoma. At final analysis, 346 deaths had occurred (165 [83.3%] in the pembrolizumab group and 181 [89.2%] in the chemotherapy group). Median OS was 8.2 months (95% CI, 6.7 to 10.3 months) and 7.1 months (95% CI, 6.1 to 8.2 months), respectively (Data Supplement); however, this difference did not reach the prespecified boundaries for significance (HR, 0.78 [95% CI, 0.63 to 0.96]; $P = .0095$; Fig 1B). The 12-month survival rate was 39.4% in the pembrolizumab group and 24.9% in the chemotherapy group (Data Supplement). The updated analysis showed that 348 deaths occurred (166 [83.8%] in the pembrolizumab group and 182 [89.7%] in the chemotherapy group). The HR was 0.77 (95% CI, 0.63 to 0.96; Data Supplement). With an additional 4 months of follow-up, the HR was 0.75 (95% CI, 0.61 to 0.93; Data Supplement). Among 401 patients with squamous cell carcinoma in the updated analysis, 167 (42%) had PD-L1 CPS ≥ 10 tumors (85 [51%] in the pembrolizumab group and 82 [49%] in the chemotherapy group). Median OS was 10.3 months (95% CI, 7.0 to 13.5 months) and 6.7 months (95% CI, 4.8 to 8.6 months), respectively (HR 0.64 [95% CI, 0.46 to 0.90]; Data Supplement).

At final analysis, 553 of all patients had died (270 [86.0%] in the pembrolizumab group and 283 [90.1%] in the chemotherapy group). Median OS was 7.1 months in each group (95% CI, [6.2 to 8.1 months] and [95% CI, 6.3 to 8.0 months], respectively; Data Supplement; HR, 0.89 [95% CI, 0.75 to 1.05]; $P = .0560$; Fig 1C). The 12-month survival rate was 32.4% with pembrolizumab and 24.2% with chemotherapy (Data Supplement).

The updated analysis showed that a total of 555 patients had died (271 [86.3%] in the pembrolizumab group and 284 [90.4%] in the chemotherapy group), with no change in median OS between groups (Data Supplement). With an additional 4 months of follow-up, the HR was 0.85 (95% CI, 0.72 to 1.10; Data Supplement). Survival outcomes were generally similar with pembrolizumab versus chemotherapy across prespecified subgroups of patients with squamous cell carcinoma and in all patients (Fig 3).

PFS

In patients with PD-L1 CPS ≥ 10 , median PFS was 2.6 months (95% CI, 2.1 to 4.1 months) in the pembrolizumab group and 3.0 months (95% CI, 2.1 to 3.7 months) in the chemotherapy group (Data Supplement; HR, 0.73 [95% CI, 0.54 to 0.97]; Fig 4A). The 12-month PFS rate was 20.8% in the pembrolizumab group and 6.7% in the chemotherapy group. Median PFS was 2.2 months (95% CI, 2.1 to 3.2 months) versus 3.1 months (95% CI, 2.2 to 3.9 months) in patients with squamous cell carcinoma (Data Supplement; HR 0.92 [95% CI, 0.75 to 1.13]; Fig 4B) and 2.1 months (95% CI, 2.1 to 2.2 months) versus 3.4 months (95% CI, 2.8 to 3.9 months) in all patients (Data Supplement; HR, 1.11 [95% CI, 0.94 to 1.31]; Fig 4C).

Tumor Response

In patients with PD-L1 CPS ≥ 10 , 23 of 107 (21.5% [95% CI, 14.1% to 30.5%]) in the pembrolizumab group and seven of 115 (6.1% [95% CI, 2.5% to 12.1%]) in the chemotherapy group had an objective response. The median duration of response was 9.3 months (range, 2.1+ to 22.6+ months) in the pembrolizumab group and 7.7 months (4.3 to 16.8+) in the chemotherapy group (Data Supplement). Antitumor activity was more favorable with pembrolizumab versus chemotherapy in patients with PD-L1 CPS ≥ 10 , regardless of histology (Data Supplement). In patients with squamous cell carcinoma, 33 of 198 (16.7% [95% CI, 11.8% to 22.6%]) in the pembrolizumab group and 15 of 203 (7.4% [95% CI, 4.2% to 11.9%]) in the chemotherapy group had an objective response (Data Supplement). In all patients, 41 of 314 (13.1% [95% CI, 9.5% to 17.3%]) in the pembrolizumab group and 21 of 314 (6.7% [95% CI, 4.2% to 10.0%]) in the chemotherapy group had an objective response (Data Supplement). Efficacy outcomes were not improved with pembrolizumab versus chemotherapy in patients with PD-L1 CPS < 10 metastatic esophageal cancer (Data Supplement).

AEs

Of 628 patients enrolled, 314 in the pembrolizumab group and 296 in the chemotherapy group received at least one dose of study treatment. The mean (standard deviation) time receiving treatment was 4.0 (5.0) months and 3.1 (2.9) months, respectively.

Patients in the chemotherapy group experienced significantly more AEs related to fatigue, diarrhea, and hematologic toxicities (Data Supplement). Treatment-related AEs occurred in 64% of patients in the pembrolizumab group and in 86% of patients in the chemotherapy group. Grade 3 or higher treatment-related AEs occurred in 18% of patients

FIG 3. Updated forest plot analysis of overall survival in subgroups of (A) patients with programmed death ligand 1 (PD-L1) combined positive score (CPS) ≥ 10 , (B) patients with squamous cell carcinoma, and (C) all patients. This analysis includes two additional survival events not included in the data at final analysis. The data cutoff date was October 15, 2018. Chemo, chemotherapy; ECOG PS, Eastern Cooperative Oncology Group performance status; Pembro, pembrolizumab.

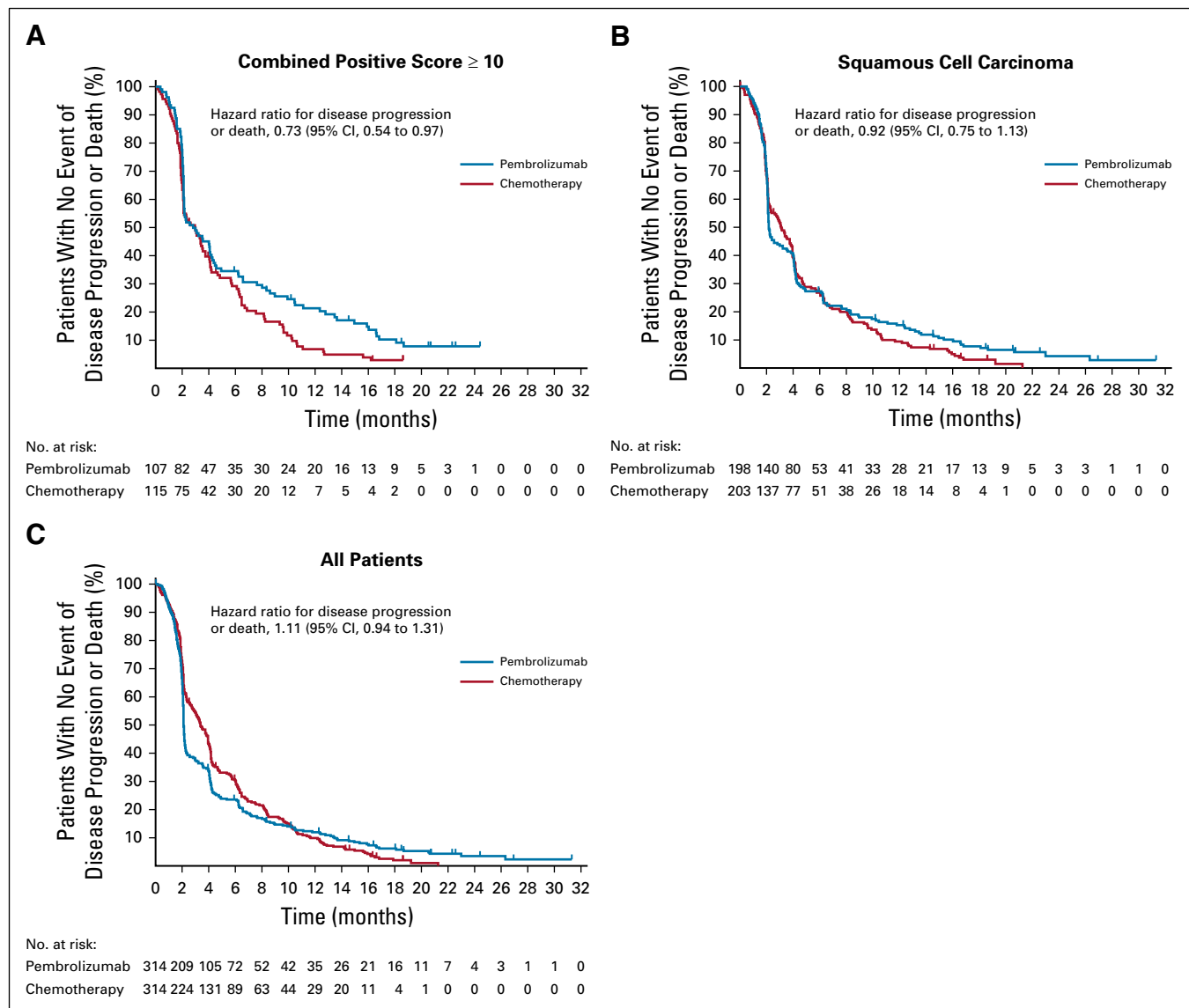


FIG 4. Progression-free survival in (A) patients with programmed death ligand 1 combined positive score ≥ 10 , (B) patients with squamous cell carcinoma, and (C) the intent-to-treat population. The data cutoff date of the original final analysis of progression-free survival was October 15, 2018.

in the pembrolizumab group and in 40.9% of patients in the chemotherapy group. Treatment-related AEs led to discontinuation in approximately 6% of patients in each group and to death for five patients in each group (Tables 2 and 3). Immune-mediated AEs and infusion reactions occurred in 23.2% of patients in the pembrolizumab group and in 7.4% of patients in the chemotherapy group (Data Supplement).

DISCUSSION

Despite recent advances, patients with metastatic esophageal cancer who progress after first-line chemotherapy continue to have limited treatment options and poor prognosis. The international, randomized phase III KEYNOTE-181 study enrolled patients worldwide and showed that

pembrolizumab provided a clinically meaningful OS benefit versus chemotherapy in a global population of patients with metastatic esophageal squamous cell carcinoma and with PD-L1 CPS ≥ 10 tumors that progressed after one prior therapy.

The survival benefit observed in patients with PD-L1 CPS ≥ 10 is highlighted by the more than twofold increase in the 12-month survival rate with pembrolizumab versus chemotherapy (43% v 20%) and by the consistent reduction in the risk of death. This consistent reduction was observed at final analysis (HR, 0.69 [95% CI, 0.52 to 0.93]; $P = .0074$), at the updated analysis (HR, 0.70 [95% CI, 0.52 to 0.94]), and with an additional 4 months of follow-up (HR, 0.67 [95% CI, 0.50 to 0.89]). A similar survival benefit was generally observed across key prespecified subgroups of

TABLE 2. Adverse Events in All Treated Patients

Adverse Event	Pembrolizumab (n = 314)	Chemotherapy (n = 296)
Any	300 (95.5)	288 (97.3)
Treatment related	202 (64.3)	255 (86.1)
Grade 3-5	57 (18.2)	121 (40.9)
Led to discontinuation	19 (6.1)	19 (6.4)
Led to death ^a	5 (1.6)	5 (1.7)

NOTE. Data are presented as No. (%). Data cutoff date was October 15, 2018.

^aGrade 5 treatment-related adverse events were myocarditis, death, and decreased WBC count in one patient each and pneumonitis in two patients in the pembrolizumab group, and pneumonia, pneumonia aspiration, sepsis, decreased neutrophil count, and hemorrhagic shock in one patient each in the chemotherapy group.

patients with PD-L1 CPS ≥ 10 , with the highest survival benefit observed in patients with squamous cell carcinoma with PD-L1 CPS ≥ 10 (HR, 0.64). In addition, Asian patients seemed to have an enhanced benefit with pembrolizumab versus chemotherapy. The reduction in the risk of death observed with pembrolizumab versus chemotherapy in the PD-L1 CPS ≥ 10 population is also supported by the clinically meaningful HR for PFS of 0.73 (95% CI, 0.54 to 0.97) and the greater than threefold increase in the 12-month PFS rate (20.8% v 6.7%). The updated analysis, which included two additional death events not in the original final analysis, missed the prespecified α level for significance (.00853) by .00002 (.00855). Together, these results show a durable benefit with pembrolizumab versus chemotherapy. Moreover, the approximately threefold increase in ORR with pembrolizumab versus chemotherapy

(21.5% v 6.1%) suggests a favorable tumor response. More patients had durable responses with pembrolizumab versus chemotherapy, with 53.5% versus 38.1% of patients, respectively, estimated to have responses with durations of ≥ 9 months. These data are consistent with those from KEYNOTE-180 that showed enrichment of efficacy outcomes in patients with PD-L1 CPS ≥ 10 metastatic esophageal cancer,¹¹ reinforcing the use of PD-L1 expression for selecting patients to be treated with pembrolizumab.

Although OS was longer with pembrolizumab versus chemotherapy in patients with squamous cell carcinoma (HR, 0.78 [95% CI, 0.63 to 0.96]; $P = .0095$), this difference did not meet the prespecified α level for significance (.0077). The survival rate at 12 months was also higher with pembrolizumab versus chemotherapy (39.4% v 24.9%). This survival benefit was generally observed across prespecified subgroups of patients with squamous cell carcinoma and was maintained at post hoc analysis and with additional follow-up, with the greatest benefit observed in patients with squamous cell carcinoma with PD-L1 CPS ≥ 10 . As in patients with PD-L1 CPS ≥ 10 , patients with squamous cell carcinoma experienced modest improvements in 12-month PFS rates and a more than twofold increase in ORR with pembrolizumab versus chemotherapy (16.7% v 7.4%). These responses were durable, with 49% of patients in the pembrolizumab group estimated to have responses with durations of at least 9 months. These data are consistent with those from the KEYNOTE-180 study that showed improved efficacy outcomes with pembrolizumab in patients with squamous cell carcinoma.¹¹ Moreover, these data are consistent with results of the phase III study of nivolumab versus chemotherapy in patients with advanced esophageal squamous cell carcinoma in the second-line setting, where a 23% reduction in the risk of death was observed with nivolumab versus chemotherapy (HR, 0.77 [95% CI, 0.62 to 0.96]; $P = .019$), with an ORR of (19% v 22%), and limited benefit in terms of risk of progression or death (HR, 1.08 [95% CI, 0.87 to 1.34]).¹²

Overall, fewer AEs were observed with pembrolizumab versus chemotherapy despite longer drug exposure (mean days receiving therapy of 112.4 v 94.8). A rainfall plot analysis showed that patients in the chemotherapy group experienced significantly more AEs related to fatigue, diarrhea, and hematologic toxicities. In addition, a lower incidence of drug-related and grade 3-5 drug-related AEs were reported with pembrolizumab versus chemotherapy. The incidence of immune-mediated AEs in the pembrolizumab group was similar to that observed previously in patients with metastatic esophageal cancer,¹⁰⁻¹² and no new safety signals were observed. The efficacy and safety outcomes observed with chemotherapy in this patient population were generally consistent with those in previous reports.⁸

A limitation of this study is the open-label nature of the study design that may have affected compliance. In addition, although OS was superior with pembrolizumab

TABLE 3. Treatment-Related Adverse Events in $\geq 10\%$ of Patients in Either Group

Adverse Event	Pembrolizumab (n = 314)		Chemotherapy (n = 296)	
	All Grades	Grade 3-5	All Grades	Grade 3-5
Fatigue	37 (11.8)	2 (0.6)	61 (20.6)	1 (0.3)
Hypothyroidism	33 (10.5)	0 (0.0)	1 (0.3)	0 (0.0)
Decreased appetite	27 (8.6)	2 (0.6)	46 (15.5)	3 (1.0)
Asthenia	22 (7.0)	4 (1.3)	34 (11.5)	3 (1.0)
Nausea	22 (7.0)	0 (0.0)	64 (21.6)	7 (2.4)
Diarrhea	17 (5.4)	2 (0.6)	60 (20.3)	9 (3.0)
Vomiting	10 (3.2)	1 (0.3)	33 (11.1)	6 (2.0)
Anemia	8 (2.5)	4 (1.3)	66 (22.3)	23 (7.8)
Alopecia	2 (0.6)	0 (0.0)	86 (29.1)	1 (0.3)
Neutrophil count decreased	2 (0.6)	1 (0.3)	50 (16.9)	29 (9.8)
Peripheral sensory neuropathy	1 (0.3)	0 (0.0)	50 (16.9)	1 (0.3)
WBC count decreased	1 (0.3)	0 (0.0)	49 (16.6)	30 (10.1)
Neutropenia	0 (0.0)	0 (0.0)	34 (11.5)	21 (7.1)

NOTE. Data are presented as No. (%). Listed are treatment-related adverse events that occurred during the study period or within 30 days thereafter (within 90 days for serious events). Data cutoff date was October 15, 2018.

compared with chemotherapy in patients with PD-L1 CPS $\geq$ 10, the study was not sufficiently powered to evaluate statistically significant differences between patients with PD-L1 CPS $\geq$ 10 squamous cell carcinoma or adenocarcinoma.

In conclusion, pembrolizumab provided a clinically meaningful survival benefit versus chemotherapy for patients with metastatic esophageal squamous cell carcinoma

and PD-L1 CPS $\geq$ 10 tumors and also in patients with metastatic esophageal squamous cell carcinoma or PD-L1 CPS $\geq$ 10 tumors, in the second line, with reduced toxicity. These data contributed to the current US Food and Drug Administration approval of pembrolizumab for patients with metastatic or locally advanced, esophageal squamous cell carcinoma with PD-L1 CPS $\geq$ 10 in the second-line setting.

AFFILIATIONS

¹National Cancer Center Hospital East, Kashiwa, Japan

²Weill Cornell Medical College, New York, NY

³Aichi Cancer Center Hospital, Nagoya, Japan

⁴CLCC Antoine Lacassagne, Nice, France

⁵IRCM, Inserm, Université Montpellier, ICM, Montpellier, France

⁶National Taiwan University Hospital, Taipei, Taiwan

⁷National Cancer Center Hospital East, Chiba, Japan

⁸University of Tsukuba Hospital, Tsukuba, Japan

⁹Asan Medical Center, Seoul, South Korea

¹⁰Samsung Medical Center, Seoul, South Korea

¹¹Institut de Cancerologie de L'Ouest, Nantes, France

¹²National Cancer Center Hospital, Tokyo, Japan

¹³Beijing Cancer Hospital, Beijing, China

¹⁴Dana Farber Cancer Institute and Harvard Medical School, Boston, MA

¹⁵PLA Cancer Centre of Nanjing Bayi Hospital, Nanjing, China

¹⁶Instituto Portugues de Oncologia Do Porto Francisco Gentil E.P.E., Porto, Portugal

¹⁷Jiangsu Cancer Hospital, Nanning, China

¹⁸Hospital de Base de Sao Jose do Rio Preto, Sao Jose do Rio Preto, Brazil

¹⁹Centre Leon Berard, Lyon, France

²⁰Centre Rene Gauducheau ICO, Saint Herblain, France

²¹University of Kansas Cancer Center, Westwood, KS

²²University Cancer Center Leipzig, Leipzig, Germany

²³Merck & Co, Inc, Kenilworth, NJ

²⁴CHU Brest – Institut de Cancerologie et d'Hematologie, Arpego Network, Brest, France

CORRESPONDING AUTHOR

Toshihiko Doi, MD, PhD, National Cancer Center Hospital East, 6-5-1, Kashiwanoha, Kashiwashi, Chiba, 277-8577, Japan; e-mail: tdoi@east.ncc.go.jp.

PRIOR PRESENTATION

Presented as an oral presentation at the 2019 Gastrointestinal Cancers Symposium (ASCO-GI), San Francisco, CA, January 17-19, 2019, and as a poster at the 2019 ASCO Annual Meeting, Chicago, IL, May 31-June 4, 2019.

SUPPORT

Supported by Grant No. NCT02564263 from Merck Sharp & Dohme, a subsidiary of Merck & Co, Inc, Kenilworth, NJ (T.K.).

CLINICAL TRIAL INFORMATION

NCT02564263

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Disclosures provided by the authors are available with this article at DOI <https://doi.org/10.1200/JCO.20.01888>.

AUTHOR CONTRIBUTIONS

Conception and design: Takashi Kojima, Chih-Hung Hsu, Toshihiko Doi, Jaafar Bennouna, Ken Kato, Lin Shen, Shu-Kui Qin, Ruixue Wang, S. Peter Kang, Jean-Philippe Metges

Administrative support: Manish A. Shah, Shu-Kui Qin

Provision of study material or patients: Takashi Kojima, Manish A. Shah, Kei Muro, Eric Francois, Antoine Adenis, Sung-Bae Kim, Ken Kato, Shu-Kui Qin, Christelle de la Fouchardiere, Helene Senellart, Florian Lordick

Collection and assembly of data: Takashi Kojima, Manish A. Shah, Kei Muro, Eric Francois, Chih-Hung Hsu, Toshihiko Doi, Toshikazu Moriwaki, Ken Kato, Lin Shen, Peter Enzinger, Shu-Kui Qin, Paula Ferreira, Jia Chen, Gustavo Giroto, Christelle de la Fouchardiere, Helene Senellart, Raed Al-Rajabi, Florian Lordick, Jean-Philippe Metges

Data analysis and interpretation: Manish A. Shah, Antoine Adenis, Chih-Hung Hsu, Toshihiko Doi, Sung-Bae Kim, Se-Hoon Lee, Jaafar Bennouna, Ken Kato, Shu-Kui Qin, Paula Ferreira, Christelle de la Fouchardiere, Florian Lordick, Ruixue Wang, Shailaja Suryawanshi, Pooja Bhagia, S. Peter Kang, Jean-Philippe Metges

Manuscript writing: All authors

Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

ACKNOWLEDGMENT

We thank the patients and their families and caregivers for participating in the study, all primary investigators and their site personnel, Luana Atherly-Henderson (MSD) for medical writing assistance, and Jonathan Cheng (MSD) for critical review.

REFERENCES

- Bray F, Ferlay J, Soerjomataram I, et al: Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 68:394-424, 2018 [Erratum: *CA Cancer J Clin* 70:313, 2020]
- Ajani JA, D'Amico TA, Bentrem DJ, et al: Esophageal and esophagogastric junction cancers, version 2.2019, NCCN Clinical Practice Guidelines in Oncology. *J Natl Compr Canc Network* 17:855-883, 2019
- Lin Y, Totsuka Y, He Y, et al: Epidemiology of esophageal cancer in Japan and China. *J Epidemiol* 23:233-242, 2013
- Shah MA: Update on metastatic gastric and esophageal cancers. *J Clin Oncol* 33:1760-1769, 2015
- Noone AM, Cronin KA, Altekruse SF, et al: Cancer incidence and survival trends by subtype using data from the Surveillance Epidemiology and End Results Program, 1992-2013. *Cancer Epidemiol Biomarkers Prev* 26:632-641, 2017

6. Muro K, Lordick F, Tsushima T, et al: Pan-Asian adapted ESMO Clinical Practice Guidelines for the management of patients with metastatic oesophageal cancer: A JSMO-ESMO initiative endorsed by CSCO, KSMO, MOS, SSO and TOS. *Ann Oncol* 30:34-43, 2019
7. Kitagawa Y, Uno T, Oyama T, et al: Esophageal cancer practice guidelines 2017 edited by the Japan Esophageal Society: Part 1. *Esophagus* 16:1-24, 2019
8. Ilson DH, van Hillegersberg R: Management of patients with adenocarcinoma or squamous cancer of the esophagus. *Gastroenterology* 154:437-451, 2018
9. Doi T, Piha-Paul SA, Jalal SI, et al: Safety and antitumor activity of the anti-programmed death-1 antibody pembrolizumab in patients with advanced esophageal carcinoma. *J Clin Oncol* 36:61-67, 2018
10. Kudo T, Hamamoto Y, Kato K, et al: Nivolumab treatment for oesophageal squamous-cell carcinoma: An open-label, multicentre, phase 2 trial. *Lancet Oncol* 18:631-639, 2017
11. Shah MA, Kojima T, Hochhauser D, et al: Efficacy and safety of pembrolizumab for heavily pretreated patients with advanced, metastatic adenocarcinoma or squamous cell carcinoma of the esophagus: The phase 2 keynote-180 study. *JAMA Oncol* 5:546-550, 2019
12. Kato K, Cho BC, Takahashi M, et al: Nivolumab versus chemotherapy in patients with advanced oesophageal squamous cell carcinoma refractory or intolerant to previous chemotherapy (ATTRACTION-3): A multicentre, randomised, open-label, phase 3 trial. *Lancet Oncol* 20:1506-1517, 2019
13. Merck Sharp & Dohme: Keytruda (pembrolizumab). https://www.merck.com/product/usa/pi_circulars/k/keytruda/keytruda_pi.pdf
14. Oken MM, Creech RH, Tormey DC, et al: Toxicity and response criteria of the Eastern Cooperative Oncology Group. *Am J Clin Oncol* 5:649-655, 1982

ASCO Answers—The Ideal Take-Home Patient Education Resource



ASCO has created helpful resources to support your patients and their caregivers. **ASCO Answers** patient education materials provide trusted information on cancer types, diagnosis, treatment, side effects, and coping in three convenient formats: fact sheets, topic-specific booklets, and comprehensive, patient-friendly guides.

ASCO Answers can be purchased from the ASCO Store at cancer.net/estore. Free domestic shipping. Members save 20%.

Cancer.Net™

Doctor-Approved Patient Information from ASCO®

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Randomized Phase III KEYNOTE-181 Study of Pembrolizumab Versus Chemotherapy in Advanced Esophageal Cancer

The following represents disclosure information provided by authors of this manuscript. All relationships are considered compensated unless otherwise noted. Relationships are self-held unless noted. I = Immediate Family Member, Inst = My Institution. Relationships may not relate to the subject matter of this manuscript. For more information about ASCO's conflict of interest policy, please refer to www.asco.org/rwc or ascopubs.org/jco/authors/author-center.

Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians (Open Payments).

Takashi Kojima

Honoraria: Oncolys BioPharma, Ono Pharmaceutical

Consulting or Advisory Role: Bristol Myers Squibb, MSD, Ono Pharmaceutical, Merck, Astellas Pharma

Research Funding: Ono Pharmaceutical (Inst), Merck Sharp & Dohme Corp (Inst), Shionogi (Inst), Astellas Amgen BioPharma (Inst), Taiho Pharmaceutical (Inst)

Manish A. Shah

Consulting or Advisory Role: Astellas Pharma, Eli Lilly Japan

Research Funding: Merck Sharp & Dohme Corp (Inst), Oncolys BioPharma (Inst), Bristol Myers Squibb (Inst)

Kei Muro

Honoraria: Takeda, Chugai Pharma, Taiho Pharmaceutical, Eli Lilly, Ono Pharmaceutical, Bayer, Sanofi, Bristol Myers Squibb

Research Funding: Ono Pharmaceutical (Inst), Merck Sharp & Dohme Corp (Inst), Daiichi Sankyo (Inst), Shionogi (Inst), Kyowa Hakko Kirin (Inst), Gilead Sciences (Inst), Merck Serono (Inst), Pfizer (Inst), Sanofi (Inst), Parexel International (Inst), Mediscience Planning (Inst), Sumitomo Dainippon (Inst), Solasia Pharma (Inst)

Eric Francois

Honoraria: Amgen, Servier, Novartis, MSD

Consulting or Advisory Role: Roche, Pierre Fabre

Research Funding: Merck Sharp & Dohme Corp (Inst)

Travel, Accommodations, Expenses: Roche, Servier

Antoine Adenis

Consulting or Advisory Role: Bristol Myers Squibb, MSD, Merck Serono

Research Funding: Sanofi (Inst), Bayer (Inst)

Travel, Accommodations, Expenses: Bristol Myers Squibb, Merck Sharp & Dohme Corp (Inst)

Chih-Hung Hsu

Honoraria: Bristol Myers Squibb, Ono Pharmaceutical, MSD, Roche

Consulting or Advisory Role: Ono Pharmaceutical, Eli Lilly, MSD, Bristol Myers Squibb, Merck Serono, Roche/Genentech

Research Funding: Ono Pharmaceutical (Inst), AstraZeneca (Inst), Merck Sharp & Dohme Corp (Inst), Genentech (Inst), Merck Serono (Inst), Taiho Pharmaceutical (Inst), Bristol Myers Squibb (Inst), BeiGene (Inst), NuCana BioMed (Inst)

Travel, Accommodations, Expenses: MSD, Roche/Genentech, Ono Pharmaceutical

Toshihiko Doi

Honoraria: Bristol Myers Squibb Japan, Ono Pharmaceutical, AbbVie, Astellas Pharma, Oncolys BioPharma, Taiho Pharmaceutical

Consulting or Advisory Role: MSD, Daiichi Sankyo, Amgen, Sumitomo Dainippon, Taiho Pharmaceutical, Takeda, AbbVie, Novartis, Bayer, Boehringer Ingelheim, Rakuten Medical

Research Funding: Taiho Pharmaceutical (Inst), Novartis (Inst), Merck Serono (Inst), Merck Sharp & Dohme Corp (Inst), Boehringer Ingelheim (Inst), Pfizer (Inst), Eli Lilly Japan (Inst), Sumitomo Group (Inst), Kyowa Hakko Kirin (Inst), Daiichi Sankyo (Inst), Bristol Myers Squibb (Inst), AbbVie (Inst), Quintiles (Inst), Eisai (Inst)

Toshikazu Moriwaki

Speakers' Bureau: Taiho Pharmaceutical, Chugai Pharma, Yakult Honsha, Takeda, Merck Serono, Sanofi, Eli Lilly Japan, Bayer Yakuhin, Ono Pharmaceutical

Research Funding: Taiho Pharmaceutical (Inst), Merck Sharp & Dohme Corp (Inst), Takeda (Inst), Yakult Honsha (Inst), Eisai (Inst)

Sung-Bae Kim

Honoraria: DAE HWA Pharmaceutical, ISU ABXIS

Consulting or Advisory Role: Eli Lilly (Inst), AstraZeneca, DAE HWA Pharmaceutical, ISU Abxis

Research Funding: Novartis (Inst), Dongkook Pharma (Inst), Genzyme (Inst), Merck Sharp & Dohme Corp (Inst)

Se-Hoon Lee

Honoraria: AstraZeneca/MedImmune, Roche, Bristol Myers Squibb, MSD

Consulting or Advisory Role: Novartis, AstraZeneca, Bristol Myers Squibb, Roche

Research Funding: Merck Sharp & Dohme Corp

Travel, Accommodations, Expenses: Novartis

Jaafar Bennouna

Honoraria: Roche, Boehringer Ingelheim, Servier, AstraZeneca, MSD, Bristol Myers Squibb, Novartis

Consulting or Advisory Role: Roche, Boehringer Ingelheim, Bristol Myers Squibb, MSD, Servier, AstraZeneca, Novartis

Research Funding: Merck Sharp & Dohme Corp (Inst)

Travel, Accommodations, Expenses: Roche, AstraZeneca, Bristol Myers Squibb

Ken Kato

Honoraria: Lilly, Bristol Myers Squibb, Ono Pharmaceutical

Consulting or Advisory Role: Ono Pharmaceutical, BeiGene, MSD, Oncolys BioPharma

Research Funding: Ono Pharmaceutical (Inst), Shionogi (Inst), Merck Sharp & Dohme Corp (Inst), Beigene (Inst)

Peter Enzinger

Consulting or Advisory Role: Merck & Co, Astellas Pharma, Taiho Pharmaceutical, Loxo, Celgene, Zymeworks, Daiichi Sankyo, AstraZeneca, Takeda

Research Funding: Merck Sharp & Dohme Corp (Inst)

Paula Ferreira

Consulting or Advisory Role: Sanofi, Merck Serono, Roche, Bristol Myers Squibb, MSD, Pierre Fabre, Novartis

Research Funding: Merck Sharp & Dohme Corp (Inst)

Speakers' Bureau: Bristol Myers Squibb

Gustavo Giroto

Honoraria: MSD

Speakers' Bureau: MSD

Research Funding: Merck Sharp & Dohme Corp (Inst), BMS Brazil (Inst)

Christelle De La Fouchardiere

Consulting or Advisory Role: Eli Lilly, Roche, Bayer, Shire, Amgen, Bristol Myers Squibb, Servier, Pierre Fabre

Research Funding: Roche, Merck Sharp & Dohme Corp (Inst)

Travel, Accommodations, Expenses: Roche, Celgene, Amgen, Bristol Myers Squibb, Amgen, Servier

Raed Al-Rajabi

Stock and Other Ownership Interests: Seattle Genetics, Actinium Pharmaceuticals

Consulting or Advisory Role: AstraZeneca, Bayer

Speakers' Bureau: Sirtex Medical

Research Funding: Eli Lilly (Inst), Merck Sharp & Dohme Corp (Inst), Sillajen (Inst), Incyte (Inst), AstraZeneca (Inst), Bristol Myers Squibb (Inst), Lexicon (Inst)

Florian Lordick

Honoraria: Eli Lilly, MSD, Bristol Myers Squibb, AstraZeneca, Elsevier, BioNTECH AG, Servier, Infomedica, Merck, Roche, Medscape

Consulting or Advisory Role: Eli Lilly, MSD, Bristol Myers Squibb, Astellas Pharma, Servier, Zymeworks, Amgen, Beigene

Research Funding: Bristol Myers Squibb (Inst), Merck Sharp & Dohme Corp (Inst)

Travel, Accommodations, Expenses: Bristol Myers Squibb, Eli Lilly

Ruixue Wang

Employment: Merck Sharp & Dohme Corp., a subsidiary of Merck & Co

Shailaja Suryawanshi

Employment: Merck Sharp & Dohme Corp., a subsidiary of Merck & Co

Stock and Other Ownership Interests: Merck Sharp & Dohme Corp., a subsidiary of Merck & Co

Pooja Bhagia

Employment: Merck Sharp & Dohme Corp., a subsidiary of Merck & Co

Stock and Other Ownership Interests: Merck Sharp & Dohme Corp., a subsidiary of Merck & Co

S. Peter Kang

Employment: Merck Sharp & Dohme Corp., a subsidiary of Merck & Co
Patents, Royalties, Other Intellectual Property: Patent related to MK 3475 stability

Jean-Philippe Metges

Honoraria: Bristol Myers Squibb, Eli Lilly, Syncore, Novartis, Sanofi
Research Funding: Merck Sharp & Dohme Corp (Inst)
Travel, Accommodations, Expenses: LEO Pharma, MSD, Amgen
No other potential conflicts of interest were reported.